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2. This question is about acids and alkalis. Lisa started filling in the following table.

Chemical name	Chemical formula	Colour with Universal Indicator	pH	Acid, alkali or neutral
sulfuric acid	H_2SO_4	red	1	acid
hydrochloric acid	HCl	red	....	acid
calcium hydroxide	$Ca(OH)_2$	purple	12	alkali
sodium hydroxide	NaOH	....	14	alkali
water	....	green	7	....

(a) Complete the table. [4]

(b) Lisa added magnesium to sulfuric acid. She recorded the temperature of the mixture during the reaction.

The word equation for this reaction is shown below.

magnesium + sulfuric acid ⟶ magnesium sulfate + hydrogen

The temperatures are shown in the table below.

Time (s)	Temperature (°C)
0	20
10	40
20	50
30	60
40	60
50	58

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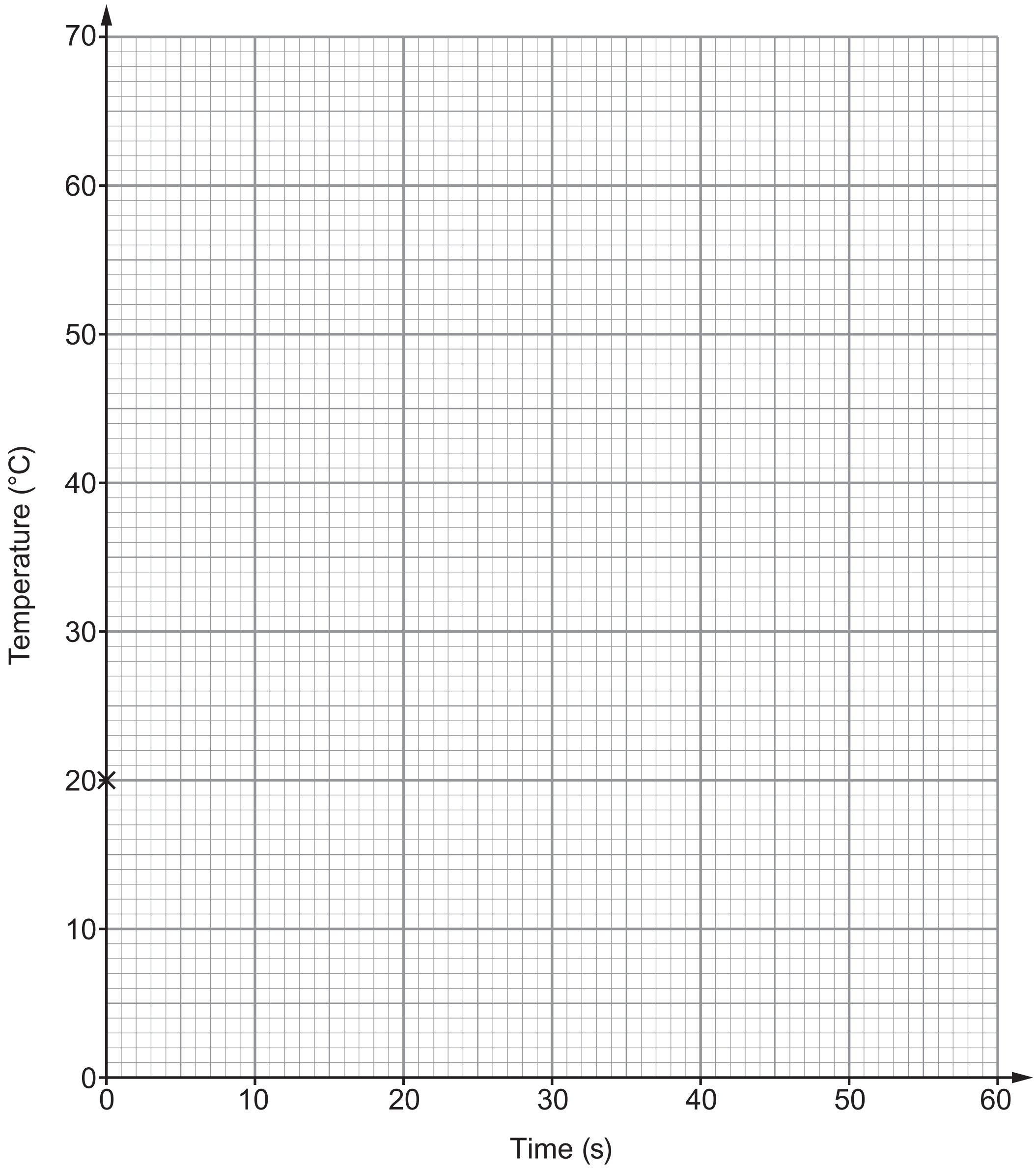
(i) Use the data in the table to plot a graph on the grid below and join the points with a ruler. The first point has been plotted for you. [3]



(ii) Calculate the temperature change during the first 20 seconds. [2]

temperature change = ... °C

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(iii) Complete the following table to describe how the temperature changes over the 50 seconds. The first two rows have been completed for you. [3]

Time (s)	Temperature decreases	Temperature stays the same	Temperature increases
0–10			✓
10–20			✓
20–30			
30–40			
40–50			

(c) Sulfuric acid has the formula H_2SO_4.

Complete the following table to calculate the relative formula mass of H_2SO_4. [3]

Element	Number of atoms	Relative atomic mass (A_r)	Total mass of the element in H_2SO_4
hydrogen	2	1	....
sulfur	1	32	....
oxygen	4	16	64
		relative formula mass (M_r)	....

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